SIEM Implementation using ELK Stack

Executive Summary

This task focuses on setting up a simple SIEM environment using the ELK stack and connecting it with a Windows machine to collect and monitor logs. Elasticsearch and Kibana were configured, and an Elastic Agent was installed and successfully enrolled to send data to the system. Sysmon was also used to generate more detailed system activity logs, especially related to process execution. After completing the setup, the agent reached a healthy state and logs started appearing in Kibana, confirming that the connection was working properly. This setup provides a clear idea of how a SIEM system can be used for monitoring and gives practical experience in collecting and analyzing logs in a real environment.

Objective

The main objective of this task is to build a simple SIEM environment using the ELK stack in order to collect and monitor logs from a Windows machine. The goal is to understand how logs are centralized and how endpoint activity can be observed through tools like Kibana. This also helps in gaining practical experience with log collection, agent configuration, and basic monitoring, which are important concepts in security operations.

System Architecture

The setup used in this task is based on the ELK stack, where each component has a specific role in handling and visualizing logs. Elasticsearch is responsible for storing and indexing the data, while Kibana is used to visualize and monitor the logs in a user-friendly way. To collect logs from the Windows machine, an Elastic Agent was installed and connected to the system through Fleet. In addition, Sysmon was used to generate detailed system activity logs, especially for process execution and system behavior. All logs collected from the endpoint are sent by the agent to Elasticsearch, and then viewed through Kibana, creating a centralized monitoring environment.

Implementation & Evidence

The following section documents the implementation process of the ELK stack and Fleet Server setup. Each step is supported with evidence to demonstrate successful configuration and operation.

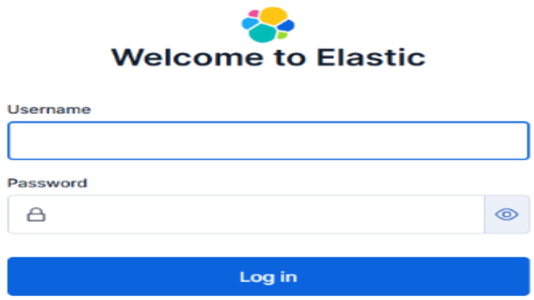

Figure 1: Initial Elastic setup process showing system configuration and service initialization.

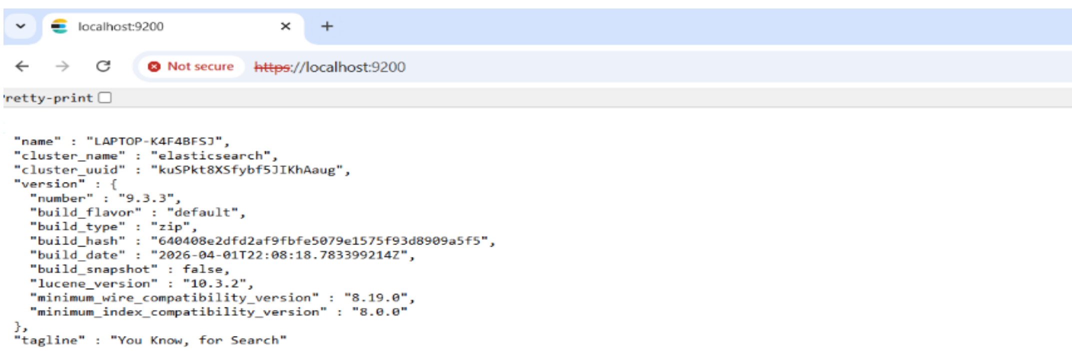

Figure 2: Elasticsearch API response confirming that the Elasticsearch service is running successfully.

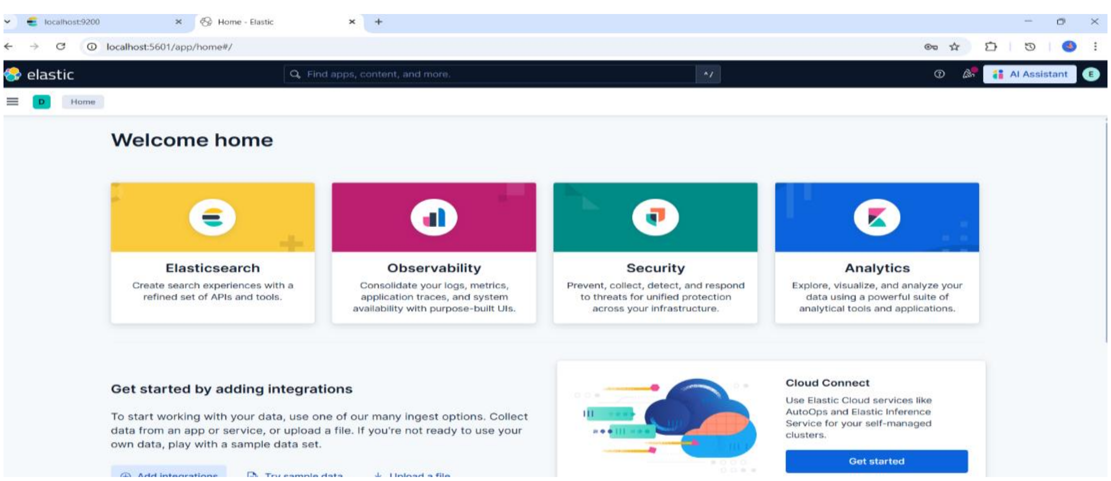

Figure 3: Kibana dashboard indicating successful access to the Elastic interface.

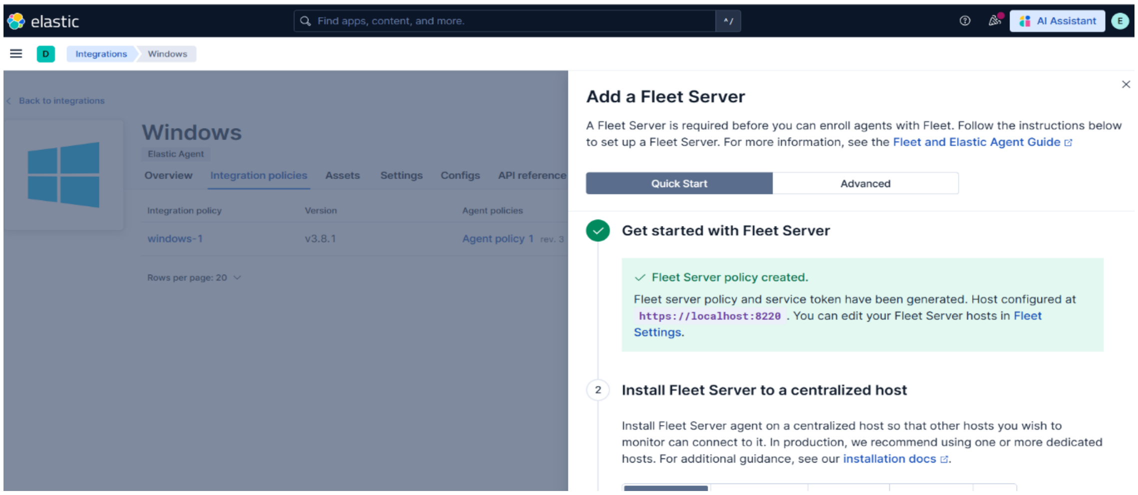

Figure 4: Fleet Server setup showing successful creation of Fleet Server policy and configuration.

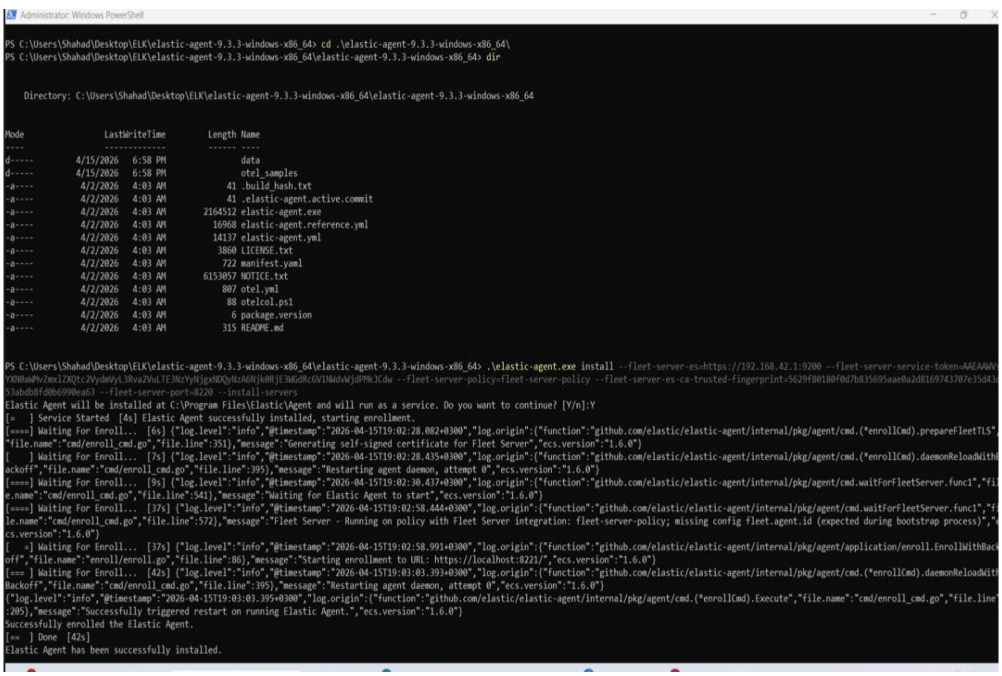

Figure 5: Elastic Agent installation and successful enrollment using PowerShell.

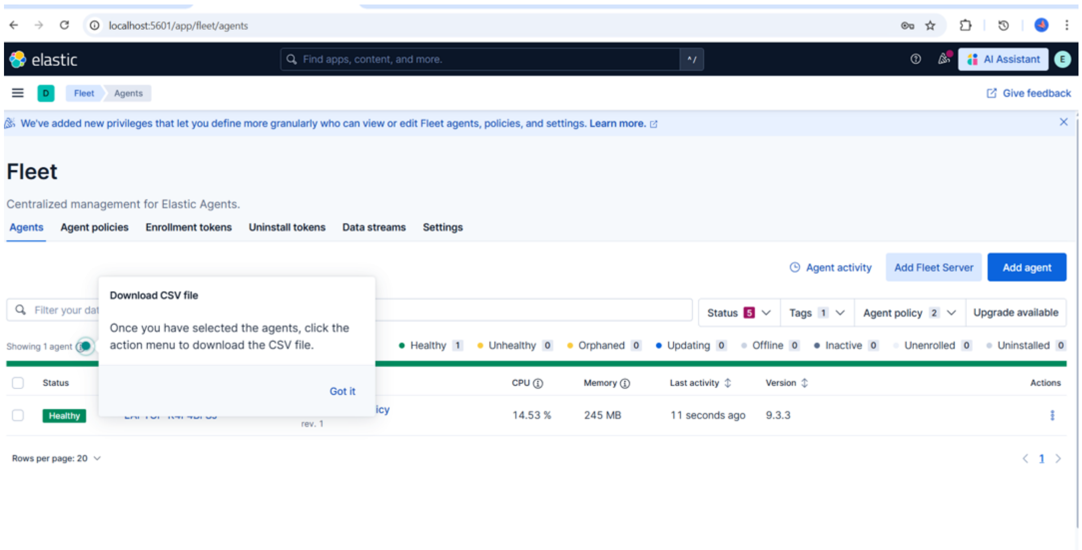

Figure 6: Agent status displaying a healthy connection between the Elastic Agent and Fleet Server.

Impact & Lessons Learned

The implementation of the ELK stack and Fleet Server provided a centralized monitoring environment, allowing better visibility and control over system activities. Through this setup, it became clear how important centralized logging and endpoint monitoring are for understanding system behavior and supporting future security analysis. The project also highlighted the importance of proper configuration, as even small misconfigurations can affect system health and agent connectivity. Additionally, the process improved understanding of how Elastic Agent communicates with Fleet Server and how data flows from endpoints to Elasticsearch for monitoring and analysis.

Recommendations & Mitigation

During the setup, some issues appeared such as agent instability and command execution problems, especially when using different environments like CMD and PowerShell. To mitigate these issues, it is recommended to use the correct terminal (PowerShell) when running Elastic commands and ensure that the correct package version (Windows) is used during installation.

For future improvements, the system can be enhanced by adding alerting rules to improve monitoring and visibility of system activity. It is also recommended to deploy the agent on multiple endpoints instead of a single machine to simulate a more realistic environment. In addition, dashboards can be configured in Kibana to provide better visualization of system activity.

Sample Logs

Sample logs were collected from the local machine using the Elastic Agent. The logs include system and application events with timestamps, demonstrating successful data collection and ingestion into Elasticsearch. These logs confirm that the monitoring setup is functioning correctly and that system activity is being tracked in real time.

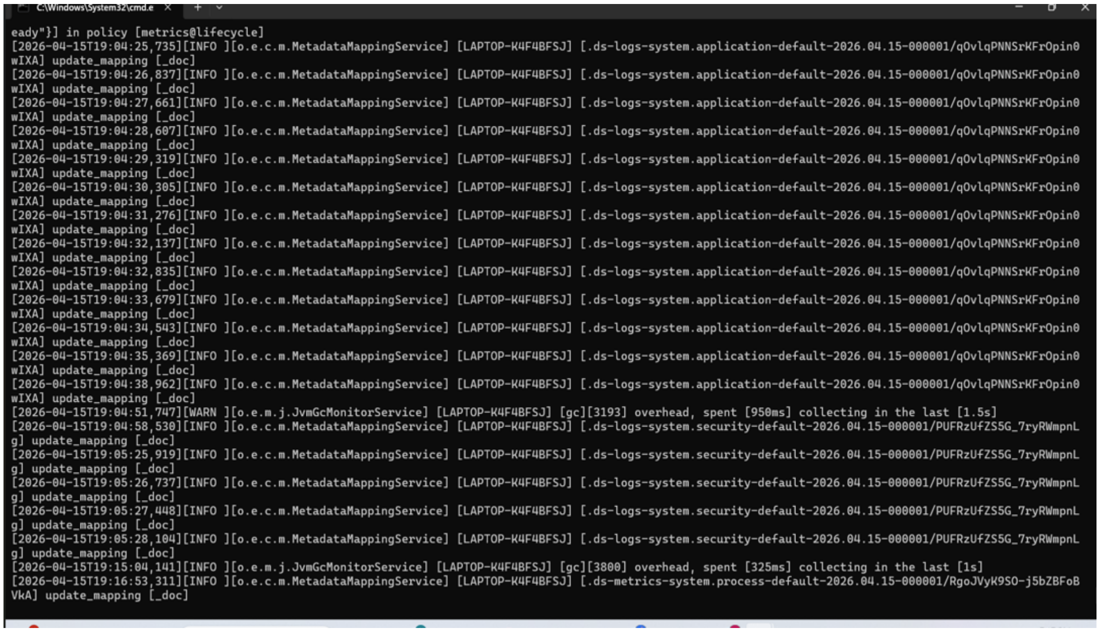

Figure 7: Sample system logs collected from the local machine, showing real-time monitoring and data ingestion.

Conclusion

In this task, a basic SIEM environment was successfully implemented using the ELK stack. The system was able to collect and display logs from a Windows machine through Elastic Agent and Kibana. Despite some minor challenges during the setup process, the overall configuration was completed successfully. This task provided a clear understanding of how SIEM systems work in practice and how logs can be used for monitoring and analysis.